Discovery Lab

Exam 2 Revision

Lower Secondary Stage 2 · Science · Lessons 7 to 9

Energy and you, fuels, and renewable resources.



Lesson 7: Energy and You

Lesson 8: Fuels

Lesson 9: Renewable Resources

Name: _____ Class: _____ Date: _____

LESSON 7

7 Energy and You

Where your energy comes from, where it goes, and why it is never used up.

Key ideas

Energy is the **ability to do work or cause change**.

Energy **cannot be created or destroyed**, only **transferred** from one store or form to another, or **stored**.

Food stores **chemical energy**. Your body transfers it to **movement** and **heat**.

The transfer is **not 100% efficient**: some energy always leaves as **heat**.

Energy you do not use can be **stored as fat**.

Food	Main energy store	Used for
Banana	Chemical	Movement and warmth
Bread	Chemical (carbohydrates)	Brain and muscle energy
Chocolate	Chemical (sugar + fat)	Quick-release energy
Water	None: not a fuel	Keeps the body hydrated

DID YOU KNOW?

Water has no calories. It supports your body but it is not an energy store.

Check yourself

- [ ] What is energy?
- [ ] What happens to energy when you run?
- [ ] Why is water not an energy store?
- [ ] Where does energy you do not use go?
- [ ] Why is the transfer not 100% efficient?

REVISION TASK

Draw an energy chain from a banana to a runner, labelling the chemical, movement and heat energy.

LESSON 8

8 Fuels

Stored chemical energy that is released as heat when burned.

Key ideas

A **fuel** stores energy and releases it as **heat** (and light) when **burned**. This is **combustion**.

Fossil fuels: coal, oil and natural gas.

Burning them releases **carbon dioxide (CO_2)**, which traps heat and contributes to **climate change**.

They are **non-renewable**: they take **millions of years** to form.

Power stations burn fuel to heat water, make steam, turn a turbine and generate **electricity**.

Fuel	Made from	Formed over
Coal	Ancient land plants	Millions of years
Oil	Ancient sea creatures	Millions of years
Natural gas	Sea creatures + pressure	Same as oil, deeper underground

DID YOU KNOW?

Over 80% of the world's energy still comes from fossil fuels.

Check yourself

- [ ] What is a fuel?
- [ ] Name three fossil fuels.
- [ ] Why are fossil fuels non-renewable?
- [ ] How does burning fuels affect the climate?
- [ ] How does a power station make electricity?

REVISION TASK

Draw the steps a power station uses to turn a fuel into electricity.

LESSON 9

9 Renewable Resources

Energy that is replaced naturally and will not run out.

Key ideas

Renewable energy is replaced naturally and will not run out.

Each source has **advantages** and **disadvantages**.

Different places suit different sources: **windy** places suit wind, **sunny** places suit solar, **rivers** suit hydroelectric.

Renewables are not used everywhere because of **location**, **cost** and **weather**.

A good long answer gives **factors** and **justifies** them with examples.

Source	Advantage	Disadvantage
Solar	Free energy from sunlight	Poor in cloud or at night
Wind	No pollution	Needs windy weather
Hydroelectric	Reliable and powerful	Dams can flood habitats
Biomass	Uses waste or crops	Still releases some CO_2
Geothermal	Works all the time	Only works in certain places

DID YOU KNOW?

In one hour the Sun gives Earth more energy than the whole world uses in a year.

Check yourself

- [ ] What does renewable mean?
- [ ] Give one advantage and one disadvantage of solar.
- [ ] Which source suits a windy coast?
- [ ] Why is biomass not completely clean?
- [ ] Why are renewables not used everywhere?

REVISION TASK

Make a table of five renewable sources with one advantage and one disadvantage of each.

FINAL PRACTICE

Final practice

Tick each box when you can answer without looking.

Final practice

- [ ] Describe the energy changes when you eat and run.
- [ ] How are fossil fuels formed?
- [ ] Why do fossil fuels cause climate change?
- [ ] Name two renewable sources and one disadvantage of each.
- [ ] What should a government consider before replacing fossil fuels?

CHALLENGE

Write a short paragraph explaining why a country should use more renewable energy, and one difficulty it might face.

Revision checklist

Revision task	Done?
Reviewed the Canvas slides for Lessons 7–9	☐
Answered every "Check yourself" question	☐
Drew energy chains and tables	☐
Practised the long answers on the website	☐
Played the labs and Keep the Lights On	☐
Asked for help with anything unclear	☐

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ANSWER KEY

Answer key

Check your own answers against these. If one is hard, go back to the lesson.

Lesson 7: Energy and You

1. The ability to do work or cause change.
2. Chemical energy from food is transferred to movement and heat.
3. It has no calories, so it is not a fuel.
4. It is stored, for example as fat.
5. Some energy always leaves as heat.

Lesson 8: Fuels

1. A substance that stores energy and releases it as heat when burned.
2. Coal, oil and natural gas.
3. They take millions of years to form and cannot be replaced quickly.
4. It releases CO_2, which traps heat.
5. Fuel heats water to steam, the steam turns a turbine and the generator makes electricity.

Lesson 9: Renewable Resources

1. Replaced naturally and will not run out.
2. Free energy; does not work well in cloud or at night.
3. Wind.
4. Burning it still releases CO_2.
5. Location, cost and weather limit them.

Final practice

1. Chemical energy from food becomes movement and heat.
2. From ancient plants and sea creatures over millions of years.
3. Burning them releases CO_2, which traps heat.
4. For example solar (needs sun) and wind (needs wind).
5. Location, cost, weather, reliability and the effect on the environment.

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You are going to do great! Use this booklet to plan your study time, then bring it all together.

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